

In sparse mode [RFC 4601], the number of routers with attached group members is small with respect to the total number of routers; group members are widely dispersed. PIM sparse mode uses rendezvous points to set up the multicast distribution tree. In **source-specific multicast (SSM)** [RFC 3569, RFC 4607], only a single sender is allowed to send traffic into the multicast tree, considerably simplifying tree construction and maintenance.

When PIM and DVMP are used within a domain, the network operator can configure IP multicast routers within the domain, in much the same way that intra-domain unicast routing protocols such as RIP, IS-IS, and OSPF can be configured. But what happens when multicast routes are needed between different domains? Is there a multicast equivalent of the inter-domain BGP protocol? The answer is (literally) yes. [RFC 4271] defines multiprotocol extensions to BGP to allow it to carry routing information for other protocols, including multicast information. The Multicast Source Discovery Protocol (MSDP) [RFC 3618, RFC 4611] can be used to connect together rendezvous points in different PIM sparse mode domains. An excellent overview of the current state of multicast routing in the Internet is [RFC 5110].

Let us close our discussion of IP multicast by noting that IP multicast has yet to take off in a big way. For interesting discussions of the Internet multicast service model and deployment issues, see [Diot 2000, Sharma 2003]. Nonetheless, in spite of the lack of widespread deployment, network-level multicast is far from “dead.” Multicast traffic has been carried for many years on Internet 2, and the networks with which it peers [Internet2 Multicast 2012]. In the United Kingdom, the BBC is engaged in trials of content distribution via IP multicast [BBC Multicast 2012]. At the same time, application-level multicast, as we saw with PPLive in Chapter 2 and in other peer-to-peer systems such as End System Multicast [Chu 2002], provides multicast distribution of content among peers using application-layer (rather than network-layer) multicast protocols. Will future multicast services be primarily implemented in the network layer (in the network core) or in the application layer (at the network’s edge)? While the current craze for content distribution via peer-to-peer approaches tips the balance in favor of application-layer multicast at least in the near-term future, progress continues to be made in IP multicast, and sometimes the race ultimately goes to the slow and steady.

4.8 Summary

In this chapter, we began our journey into the network core. We learned that the network layer involves each and every host and router in the network. Because of this, network-layer protocols are among the most challenging in the protocol stack.

We learned that a router may need to process millions of flows of packets between different source-destination pairs at the same time. To permit a router to process such a large number of flows, network designers have learned over the years that the router’s tasks should be as simple as possible. Many measures can be taken

to make the router's job easier, including using a datagram network layer rather than a virtual-circuit network layer, using a streamlined and fixed-sized header (as in IPv6), eliminating fragmentation (also done in IPv6), and providing the one and only best-effort service. Perhaps the most important trick here is *not* to keep track of individual flows, but instead base routing decisions solely on hierarchically structured destination addresses in the datagrams. It is interesting to note that the postal service has been using this approach for many years.

In this chapter, we also looked at the underlying principles of routing algorithms. We learned how routing algorithms abstract the computer network to a graph with nodes and links. With this abstraction, we can exploit the rich theory of shortest-path routing in graphs, which has been developed over the past 40 years in the operations research and algorithms communities. We saw that there are two broad approaches: a centralized (global) approach, in which each node obtains a complete map of the network and independently applies a shortest-path routing algorithm; and a decentralized approach, in which individual nodes have only a partial picture of the entire network, yet the nodes work together to deliver packets along the shortest routes. We also studied how hierarchy is used to deal with the problem of scale by partitioning large networks into independent administrative domains called autonomous systems (ASs). Each AS independently routes its datagrams through the AS, just as each country independently routes its postal mail through the country. We learned how centralized, decentralized, and hierarchical approaches are embodied in the principal routing protocols in the Internet: RIP, OSPF, and BGP. We concluded our study of routing algorithms by considering broadcast and multicast routing.

Having completed our study of the network layer, our journey now takes us one step further down the protocol stack, namely, to the link layer. Like the network layer, the link layer is also part of the network core. But we will see in the next chapter that the link layer has the much more localized task of moving packets between nodes on the same link or LAN. Although this task may appear on the surface to be trivial compared with that of the network layer's tasks, we will see that the link layer involves a number of important and fascinating issues that can keep us busy for a long time.



Homework Problems and Questions

Chapter 4 Review Questions

SECTIONS 4.1–4.2

- R1. Let's review some of the terminology used in this textbook. Recall that the name of a transport-layer packet is *segment* and that the name of a link-layer packet is *frame*. What is the name of a network-layer packet? Recall that both routers and link-layer switches are called *packet switches*. What is the fundamental difference between a router and link-layer switch? Recall that we use the term *routers* for both datagram networks and VC networks.